

KINEMATICS

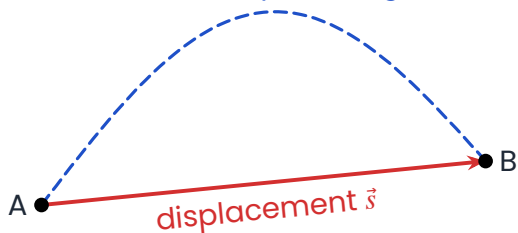
Motion in a Line • Basics

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1 Distance vs Displacement

$$\vec{s} = \vec{r}_2 - \vec{r}_1, \quad |\vec{s}| \leq \text{distance}$$

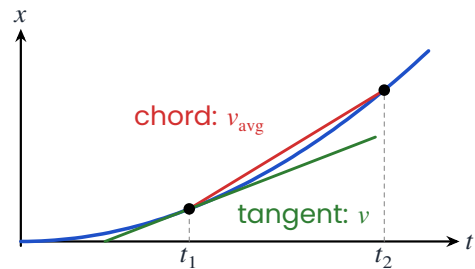
distance = path length



Distance: a scalar, never decreases.
Displacement: a vector, zero for a round trip.

2 Average vs Instantaneous

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \quad v = \frac{dx}{dt}$$

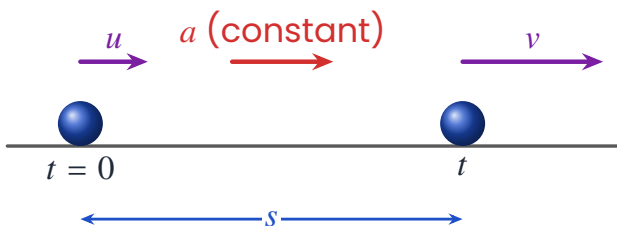


Average speed = distance/time,
which can differ from $|v_{\text{avg}}|$.

3 Equations of Motion

$$v = u + at, \quad s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

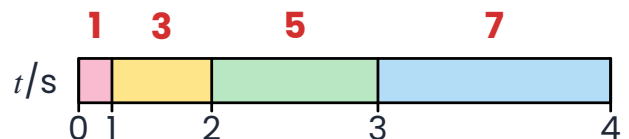


Constant a only. If a varies, integrate
 $a = dv/dt = v dv/dx$.

4 Displacement in the nth Second

$$s_n = u + \frac{a}{2} (2n - 1)$$

metres in each second, from rest



Constant a . Its unit is metre, though it
looks like m/s.

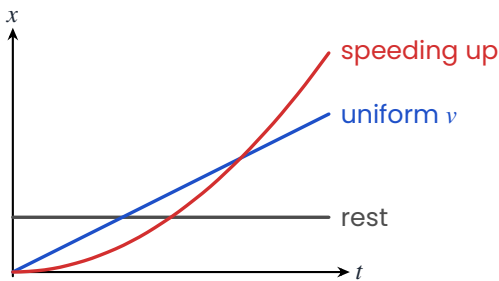
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Graphs • Free Fall

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5 x-t Graph: Slope = Velocity

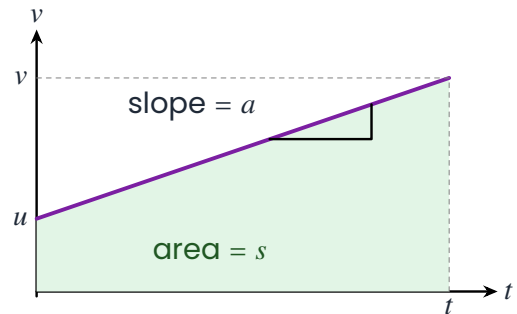
$$v = \frac{dx}{dt} = \text{slope of } x-t$$



Never vertical (infinite v) and never turns back in time.

6 v-t Graph: Slope and Area

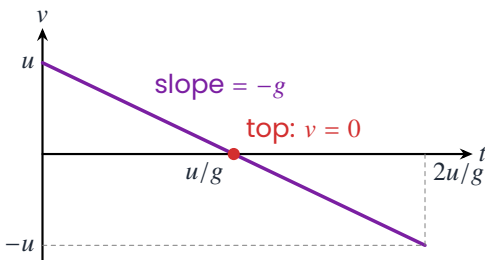
$$a = \text{slope}, \quad s = \text{area}$$



Area below the t -axis is negative; distance adds the sizes of all areas.

7 Thrown Up • Free Fall

$$t_{\text{up}} = \frac{u}{g}, \quad H = \frac{u^2}{2g}$$



Uniform g , no air resistance. Top: $v = 0$,
 $a = g$. Dropped: $v = gt$, $h = gt^2/2$.

8 Quick Numerical: Free Fall

A ball is dropped from 80 m ($g = 10 \text{ m s}^{-2}$). Find the time to fall, the landing speed and the distance in the last second.

$$t = \sqrt{2h/g} = 4 \text{ s}$$

$$v = gt = 40 \text{ m/s}$$

$$s_4 = \frac{g}{2}(2 \cdot 4 - 1) = 35 \text{ m}$$

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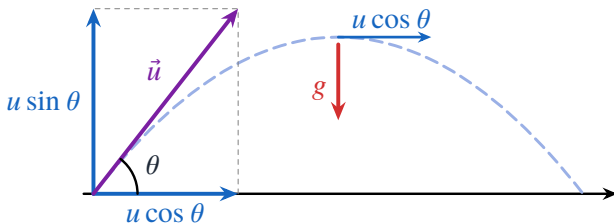
Projectile • The Basics

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9 Split into Components

$$u_x = u \cos \theta, \quad a_x = 0$$

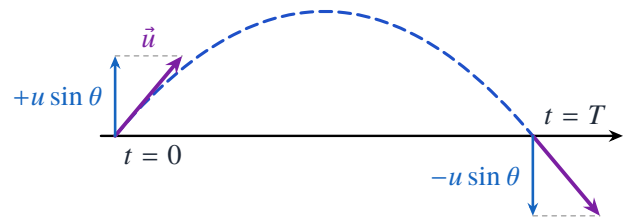
$$u_y = u \sin \theta, \quad a_y = -g$$



No air resistance, uniform g : x and y are independent; v_x never changes.

10 Time of Flight

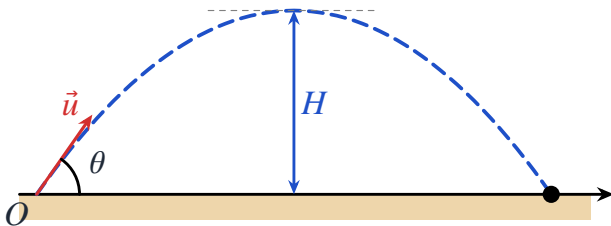
$$T = \frac{2u \sin \theta}{g}$$



Level ground, no air resistance, uniform g . Up time = down time.

11 Maximum Height

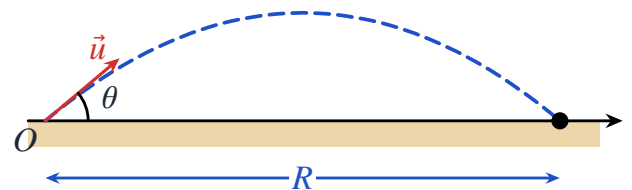
$$H = \frac{u^2 \sin^2 \theta}{2g}$$



No air resistance, uniform g . At the top $v_y = 0$, $v = u \cos \theta$, at $t = T/2$.

12 Horizontal Range

$$R = \frac{u^2 \sin 2\theta}{g} = u \cos \theta \cdot T$$



Level ground (lands at launch height), no air resistance, uniform g .

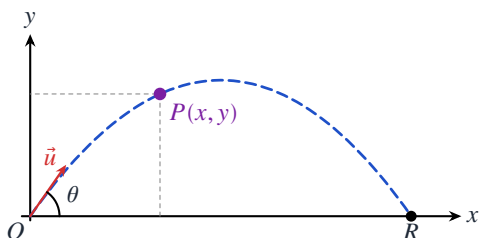
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Projectile • Path and Range

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13 Equation of Trajectory

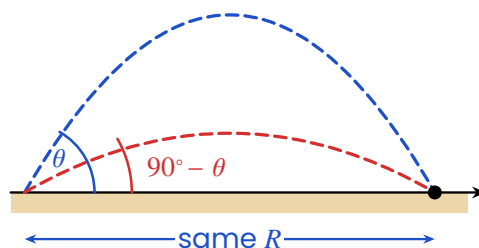
$$y = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}$$



No air resistance, uniform g . A parabola: $y = x \tan \theta (1 - x/R)$.

14 Complementary Angles

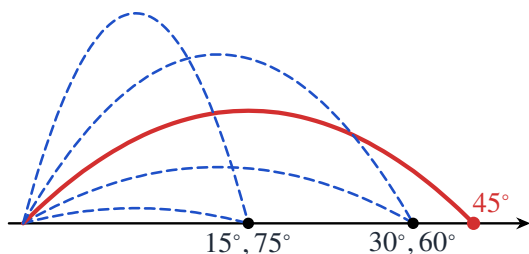
$$R_{\theta} = R_{90^{\circ}-\theta}, \quad T_1 T_2 = \frac{2R}{g}$$



Same u , level ground, no air resistance, uniform g . $H_1 + H_2 = u^2/2g$.

15 Maximum Range at 45°

$$R_{\max} = \frac{u^2}{g} \quad (\theta = 45^{\circ})$$



Level ground, no air resistance, uniform g ; $H = R_{\max}/4$. Not best from a height or on an incline.

16 Quick Numerical: Projectile

20 m/s at 30° , level ground, no air resistance, $g = 10 \text{ m s}^{-2}$. Find T , H , R .

$$T = \frac{2u \sin \theta}{g} = 2 \text{ s}$$

$$H = \frac{u^2 \sin^2 \theta}{2g} = 5 \text{ m}$$

$$R = \frac{u^2 \sin 2\theta}{g} = 34.6 \text{ m}$$

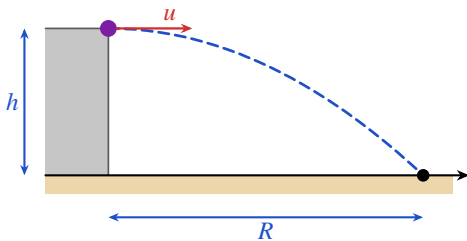
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Projectile • Advanced Cases

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17 Thrown from a Height

$$t = \sqrt{\frac{2h}{g}}, \quad R = u \sqrt{\frac{2h}{g}}$$

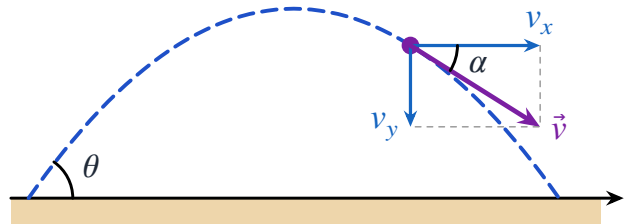


No air resistance, uniform g . $u_y = 0$, so t depends on h only.

18 Velocity at Time t

$$v_x = u \cos \theta, \quad v_y = u \sin \theta - gt$$

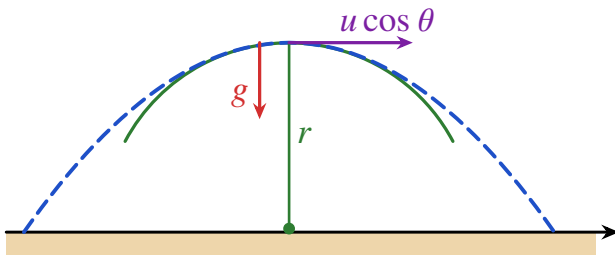
$$\tan \alpha = v_y / v_x$$



No air resistance, uniform g . $\vec{v}$ is along the tangent; $\alpha < 0$ on the way down.

19 Radius of Curvature at the Top

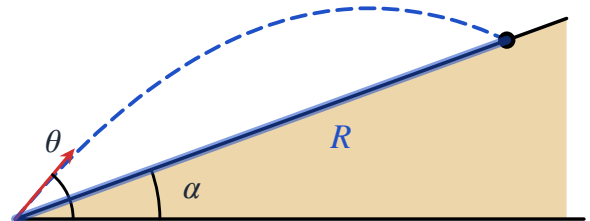
$$r = \frac{v^2}{a_{\perp}} \Rightarrow r_{\text{top}} = \frac{u^2 \cos^2 \theta}{g}$$



No air resistance, uniform g . At the top all of g is normal to $\vec{v}$; r is least.

20 Range up an Incline

$$R = \frac{2u^2 \cos \theta \sin(\theta - \alpha)}{g \cos^2 \alpha}$$



No air resistance, uniform g ; θ from horizontal. Best $\theta = 45^\circ + \alpha/2$:

$$R_{\text{max}} = u^2 / [g(1 + \sin \alpha)].$$

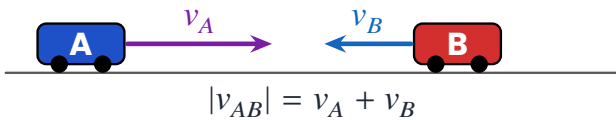
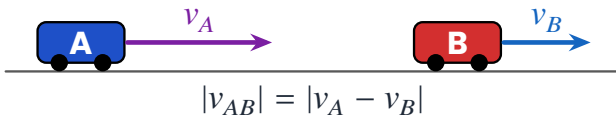
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Relative Velocity

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21 Relative Velocity in 1-D

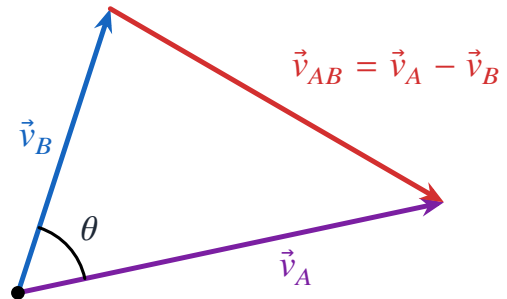
$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$



Velocity of A as seen by B. Same way:
subtract; opposite: add.

22 Relative Velocity in 2-D

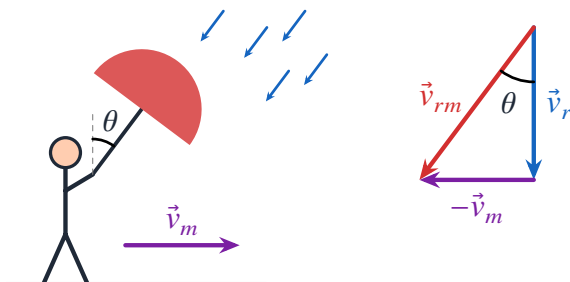
$$v_{AB} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos \theta}$$



θ = angle between $\vec{v}_A$ and $\vec{v}_B$;
 $\vec{v}_{BA} = -\vec{v}_{AB}$

23 Rain and Umbrella

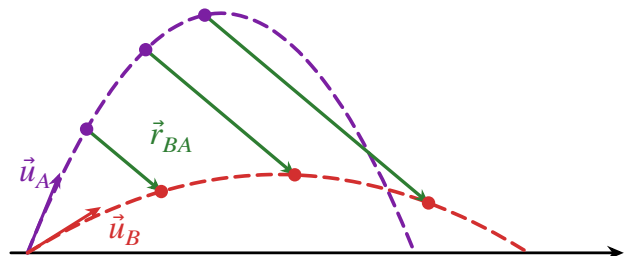
$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m, \quad \tan \theta = \frac{v_m}{v_r}$$



Rain falls vertically. Tilt the umbrella
forward by θ from the vertical.

24 Two Projectiles

$$\vec{a}_{AB} = \vec{g} - \vec{g} = 0$$



So $\vec{v}_{AB}$ is constant while both are in
flight (no air resistance): each sees
the other move in a straight line.

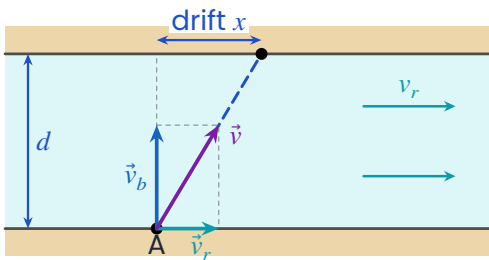
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River • Boat • Wind

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25 River: Shortest Time

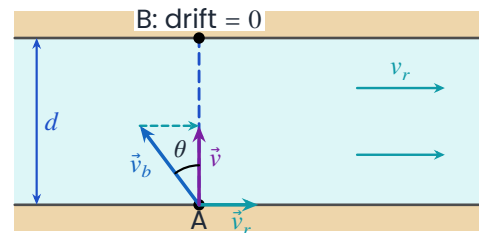
$$t_{\min} = \frac{d}{v_b}, \quad x = \frac{v_r d}{v_b}$$



Head straight across ($\vec{v}_b \perp$ bank). The current only adds the drift x .

26 River: Shortest Path

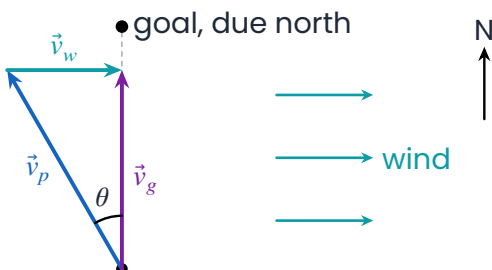
$$\sin \theta = \frac{v_r}{v_b}, \quad t = \frac{d}{\sqrt{v_b^2 - v_r^2}}$$



Zero drift needs $v_b > v_r$. Head upstream at θ from the normal.

27 Aircraft in Wind

$$\sin \theta = \frac{v_w}{v_p}, \quad v_g = \sqrt{v_p^2 - v_w^2}$$



Needs $v_p > v_w$. Wind $\perp$ the goal; $\vec{v}_p$ is relative to air, $\vec{v}_g = \vec{v}_p + \vec{v}_w$.

28 Quick Numerical: River

River 120 m wide, current 3 m/s, boat 5 m/s in still water. Least time and drift? Time with zero drift?

$$t_{\min} = 120/5 = 24 \text{ s}$$

$$x = 3 \times 24 = 72 \text{ m}$$

$$\sin \theta = 3/5, \quad \theta \approx 37^\circ \text{ upstream}$$

$$t = \frac{120}{\sqrt{5^2 - 3^2}} = 30 \text{ s}$$

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Quick Revision • Traps

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29 Every Result and When It Holds

Quantity	Result	Only if
v, s, v^2	$u + at, ut + \frac{1}{2}at^2, u^2 + 2as$	a constant
n th second	$s_n = u + \frac{a}{2}(2n - 1)$	a constant
T, H, R	$2u \sin \theta / g, u^2 \sin^2 \theta / 2g, u^2 \sin 2\theta / g$	level ground, no air
$R_{\max}$	u^2 / g at 45°	level ground, no air
From height h	$t = \sqrt{2h/g}, R = u\sqrt{2h/g}$	thrown horizontally, no air
Top of path	$v = u \cos \theta, r = u^2 \cos^2 \theta / g$	uniform g , no air
River, least time	$t = d/v_b, x = v_r d / v_b$	head across
River, zero drift	$\sin \theta = v_r / v_b, t = d / \sqrt{v_b^2 - v_r^2}$	$v_b > v_r$
Rain-man	$\tan \theta = v_m / v_r$	rain vertical

30 Common Traps

- At the top $v_y = 0$, but $a = g$, never 0.
- $|\text{displacement}| \leq \text{distance}$; equal only for straight-line motion with no reversal.
- Least time $\neq$ least path when crossing a river.
- 45° gives $R_{\max}$ on level ground, no air.
- s_n is in metres, not m/s.

31 Constants & Units

Quantity	Value
g near Earth's surface	$9.8 \text{ m s}^{-2} (\approx 10)$
1 km h^{-1}	$\frac{5}{18} \text{ m s}^{-1}$
$\sin 37^\circ, \cos 37^\circ$	$\approx 3/5, 4/5$
$\sin 30^\circ, \sin 60^\circ$	$1/2, \sqrt{3}/2$
v, a	$\text{m s}^{-1}, \text{m s}^{-2}$

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